

# HANYU CHENG

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## EDUCATION

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**Georgia Institute of Technology**, H. Milton Stewart School of Industrial and Systems Engineering  
Incoming Ph.D. Student in Industrial Engineering, starting Jan 2027

**Tsinghua University**, Department of Industrial Engineering Sep 2021 – Jul 2026  
Bachelor of Engineering, GPA: 3.8/4.0  
Thesis: Optimizing Urban Electric Vehicle (EV) Battery Swapping and Charging Systems  
Supervisor: Prof. Wei Qi

## RESEARCH INTERESTS

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Stochastic Control, Game Theory, Logistics.

## PUBLICATIONS

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**Cheng, H.**, Cheng, L., & Bai, X. (2026). Dynamic Carbon Intensity Indicator (CII) Management in Stochastic Tramp Shipping Market. *Transportation Research Part A: Policy and Practice*, 210, 105059. <https://doi.org/10.1016/j.tra.2026.105059>

Ai, W.<sup>\*</sup>, **Cheng, H.**<sup>\*</sup>, & Qi, W. (2026). Optimizing Urban Electric Vehicle Charging and Battery Swapping Infrastructure: A Location-Inventory-Grid Model. *Transportation Research Part E: Logistics and Transportation Review*, 214, 104897. <https://doi.org/10.1016/j.tre.2026.104897>

**Cheng, H.**, Tian, Q., & Wang, D.Z. (2025). Adding Runs of Modular Autonomous Vehicles with Skip-Stop Strategy Using Deep Reinforcement Learning. *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 4066-4073. <https://doi.org/10.1109/ITSC60802.2025.11423421>

<sup>\*</sup>These authors contributed equally.

## WORKING PAPERS

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**Cheng H.**, & Yao, D. D. (2026). Optimal Control of a Non-Stationary Inventory-Queue, with Applications to EV-Battery Swapping. <https://arxiv.org/abs/2609.35153>

Gao, S., **Cheng, H.**, Yan, C., & Jiang, G. (2026). A Markovian Traffic Equilibrium Model for Ride-Hailing. Major Revision at *Transportation Science*. <https://doi.org/10.48550/arXiv.2604.21359>

## RESEARCH APPOINTMENTS

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**Staff Associate I, Columbia University** Nov 2026 – Jan 2027  
Supervisor: Prof. David D. Yao

**Visiting Student Intern (VSI), Columbia University** Mar 2026 – May 2026  
Supervisor: Prof. David D. Yao

**Visiting Student Researcher (VSR), University of California, Berkeley** Jul 2025 – Sep 2025  
Supervisor: Prof. Chiwei Yan

## REFeree ACTIVITY

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IEEE International Conference on Intelligent Transportation Systems (ITSC), IEEE Intelligent Vehicles Symposium (IV).

**AWARDS AND SCHOLARSHIPS**

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|-----------------------------------------------------------|-----------|
| Innovation Excellence Scholarship, Tsinghua University    | 2024,2025 |
| Comprehensive Excellence Scholarship, Tsinghua University | 2022,2023 |
| Outstanding Student Leader, Tsinghua University           | 2022      |

**SKILLS**

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|----------------------|------------------------|
| <b>Programming</b>   | Python, MATLAB, R, SQL |
| <b>Miscellaneous</b> | MS Series, LaTeX       |

**LANGUAGES**

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|------------------------------------------------------------|
| Mandarin Chinese (Native), English (Fluent; TOEFL 108/120) |
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